a. Factors required for good lighting scheme

The following factors are required to be considered while designing a lighting scheme

i. Illumination level
ii. Quality of light
iii. Coefficient of utilization
iv. Depreciation factor
v. Space height ratio

b. Advantages of electrical lighting

i. Cleanliness
ii. Easy to control
iii. Economical
iv. Easy to handle
v. Steady output
vi. Better reliability
vii. Suitable for almost all purposes etc.

C. Type of work recommended and illumination level

Places	Illumination Level
Offices	100-400 lumens/ meter square
Schools	250-400 lumens/ meter square
Industry	1000 lumens/ meter square
Shops	250-500 lumens/ meter square
Hotels	80-100 lumens/ meter square
Hospitals	250-3500 lumens/ meter square

Points to remember

- The unit of solid angle is steradians
- The unit of plane angle is radians
- The unit of luminous flux is lumen
- In LED lamps, 15 watts = 900 lumens
- Glare causes annoyance, discomfort or interference with vision or eye fatigue.
- Law of illumination
  - i. Inverse square law
  - ii. Lambert's cosine law
- Carbon arc lamp is most commonly used arc lamp.
- Types of incandescent lamp
  - i. Vacuum lamp
  - ii. Gas filled lamp

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